

A taxonomy of separation

From Flory–Huggins to non-local gradients

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Why does the CalcVar community exist?

- Collaboration
- Friends
- De Giorgi
- Career
- Roermond
- Fun
- Italians 🍆

The real reason: the Modica–Mortola functional.

The Modica–Mortola functional

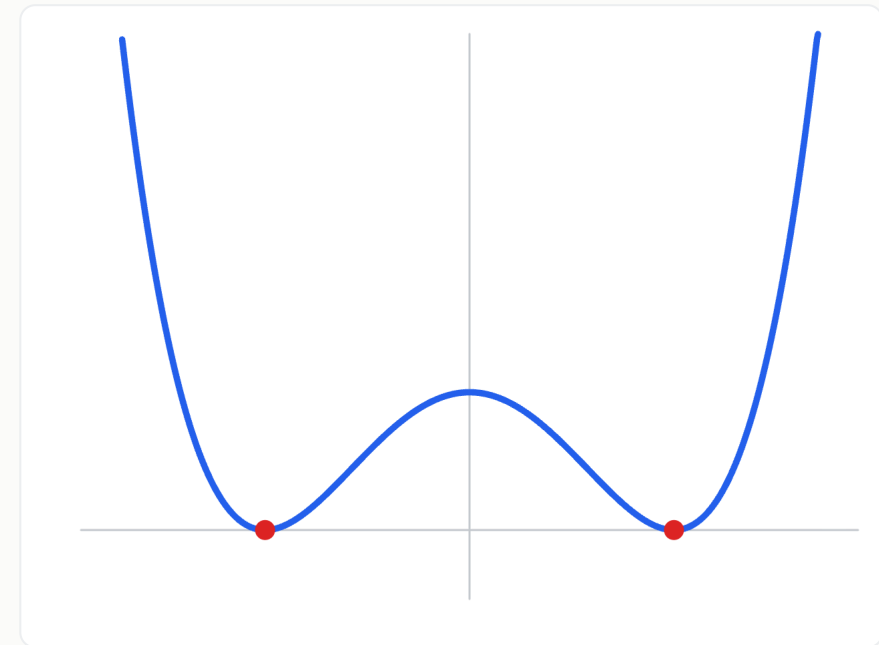
$$F_\varepsilon(u) = \int_{\Omega} [W(u) + \varepsilon^2 |\nabla u|^2] \, dx$$

with a double-well potential $W : \mathbb{R} \rightarrow [0, +\infty)$. ^[1,2]

Ubiquitous in mathematical physics:

- Cahn–Hilliard / Allen–Cahn
- Ginzburg–Landau
- Mumford–Shah

But **why** this energy? $\rightsquigarrow$ thermodynamics.



Energy minimization

The principle of ~~least~~ **stationary** action.

Translated to thermodynamics, it minimizes the

Gibbs free energy

$$\Delta F = \Delta H - T \Delta S \quad (\text{constant } P, T).$$

	fix pressure	fix volume
experimentally	good	bad
mathematically	bad	good

⇒ **work with the Helmholtz free energy.**

Helmholtz free energy

$$\Delta F = \Delta U - T \Delta S \quad (\text{constant } V, T).$$

Incompressibility gives $\Delta V = 0$; since $\Delta H = \Delta U + P \Delta V$,

$$\Delta F = \Delta H = \Delta G. \quad \checkmark$$

Roadmap — define the lattice $\rightarrow$ define the interactions $\rightarrow$ compute $\Delta F \rightarrow$ win.

The lattice model

$\Omega \subset \mathbb{R}^d$ bounded, connected, Lipschitz. For $\delta > 0$,

$$\Lambda_\delta := \delta\mathbb{Z}^d \cap \Omega, \quad N = \#\Lambda_\delta \approx |\Omega|/\delta^d.$$

Binary mixture A, B . Configuration

$$\sigma : \Lambda_\delta \rightarrow \{0, 1\}, \quad \sigma(x) = \begin{cases} 1 & A \text{ at } x \\ 0 & B \text{ at } x \end{cases}$$

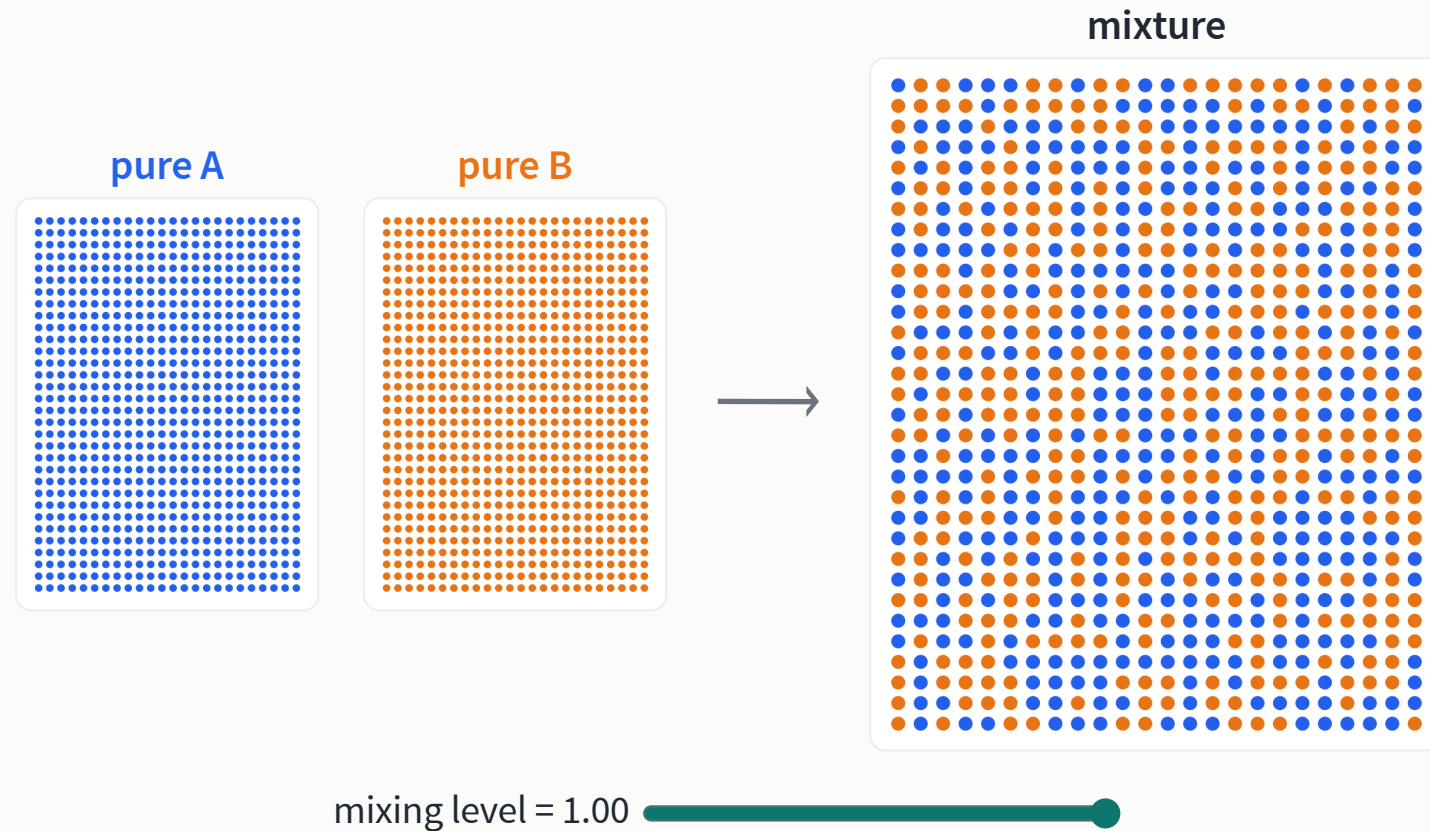
Incompressibility: $\sigma_B = 1 - \sigma_A$.

Global fractions

$$\varphi_A = \frac{N_A}{N}, \quad \varphi_B = 1 - \varphi_A.$$

Order parameter: the fraction $\varphi_A \in [0, 1]$.

Energy of mixing



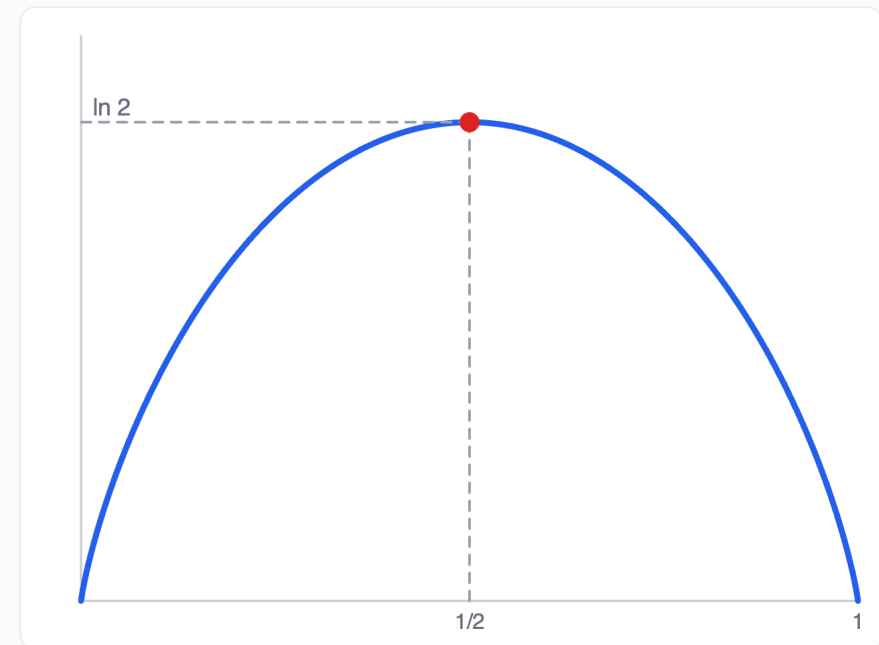
Two competing effects set the equilibrium: **entropy** wants to mix, **energy** may not.

First ingredient: entropy

Boltzmann counting of configurations gives the entropy of mixing per site

$$\Delta S = -k_B [\varphi_A \ln \varphi_A + \varphi_B \ln \varphi_B].$$

Always $\Delta S > 0$: **entropy** opposes separation.



$\Delta S(\varphi) > 0$ on $(0, 1)$.

Second ingredient: internal energy

Pairwise nearest-neighbour, coordination number z . **Mean field:** a neighbour is A with probability φ_A .

$$U_0 = \frac{z}{2} N [U_{AA} \varphi_A + U_{BB} \varphi_B],$$

$$U = \frac{z}{2} N [U_{AA} \varphi_A^2 + 2U_{AB} \varphi_A \varphi_B + U_{BB} \varphi_B^2].$$

Per site, $\Delta U = \frac{z}{2} (2U_{AB} - U_{AA} - U_{BB}) \varphi_A \varphi_B$.

Flory-Huggins interaction parameter ^[3]

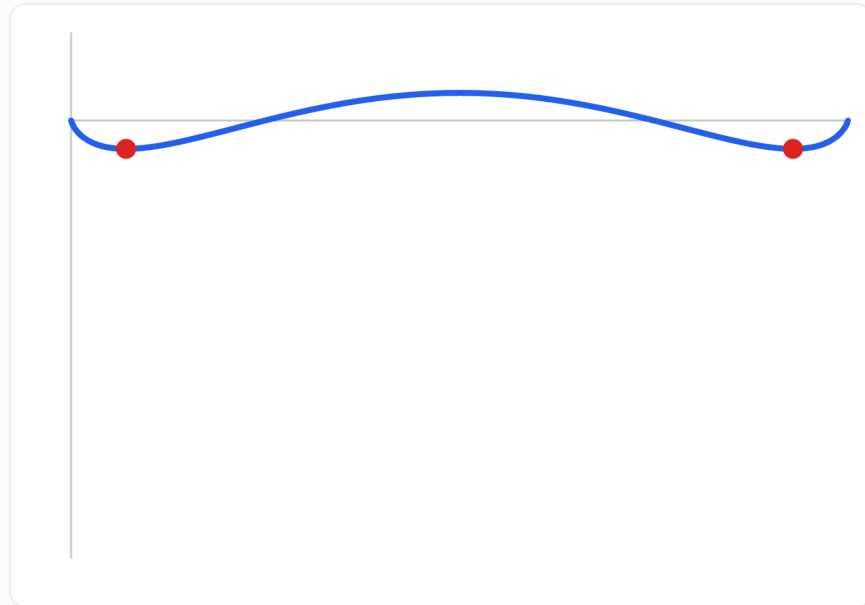
$$\chi = \frac{z}{2} \frac{2U_{AB} - U_{AA} - U_{BB}}{k_B T}$$

$$\Rightarrow \Delta U = k_B T \chi \varphi_A \varphi_B.$$

The Flory–Huggins free energy

$$\Delta F = k_B T [\varphi_A \ln \varphi_A + \varphi_B \ln \varphi_B + \chi \varphi_A \varphi_B].$$

χ controls the landscape: one well, then **two** at $\chi_c = 2$.

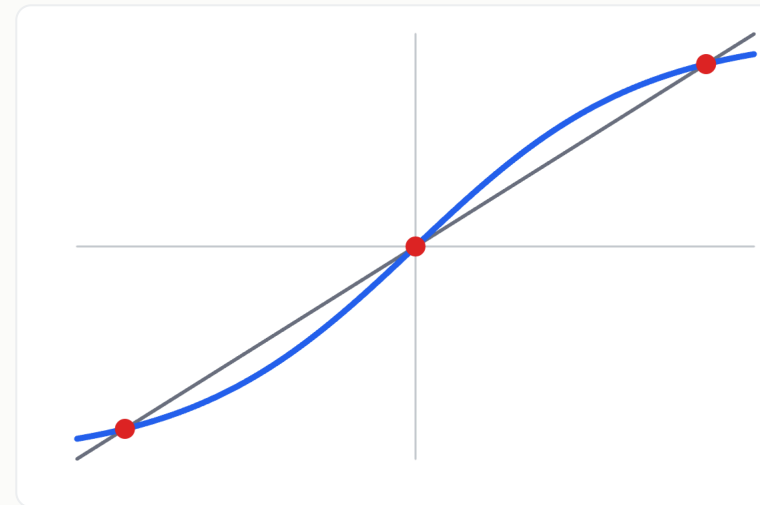


$\chi = 3.00$

Equivalently, a mean-field **Ising** lattice: separation appears iff

$$m = \tanh\left(\frac{\chi}{2} m\right)$$

has a non-zero root, i.e. $\chi > 2$ (Curie–Weiss). ^[4]



$\chi = 3.00$

Excess above equilibrium

The object that matters is the **excess** $W \approx F - \min F$ — the Modica–Mortola well.

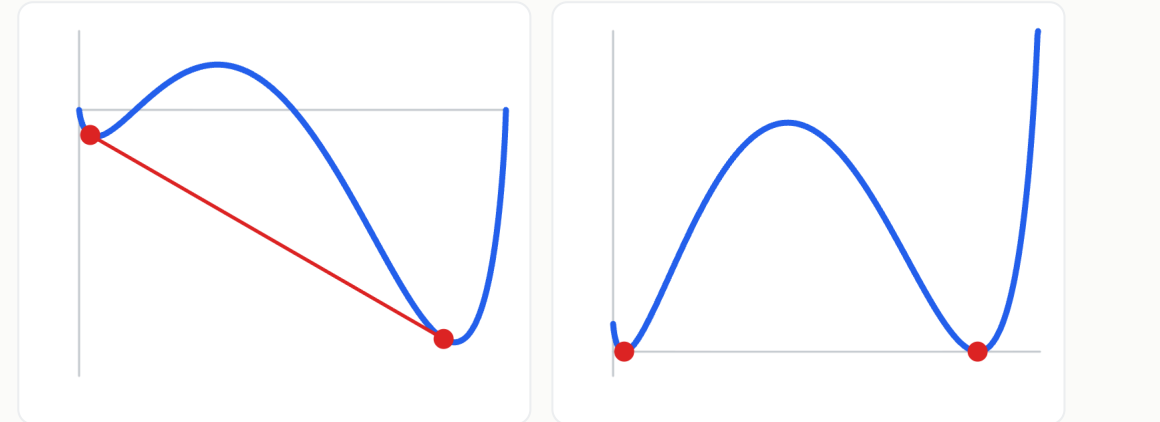
For chains of length n_A, n_B ,

$$F = \frac{\varphi_A}{n_A} \ln \varphi_A + \frac{\varphi_B}{n_B} \ln \varphi_B + \chi \varphi_A \varphi_B,$$

with coexistence set by the **common tangent**

$$F'(\varphi_1^*) = F'(\varphi_2^*) = \mu.$$

Minima of $F \neq$ minima of W unless $\mu = 0$ (monomers).

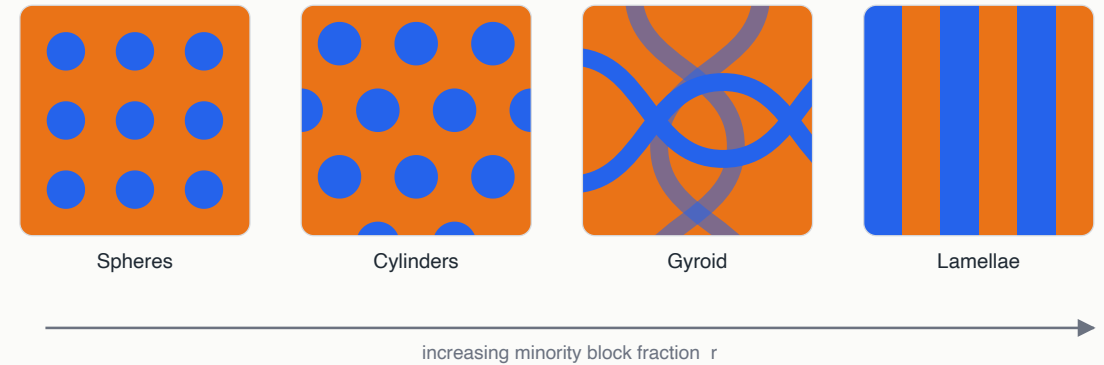


Effective parameter & block copolymers

- water-oil: $\chi_c \approx 2$, $\chi_{\text{exp}} \approx 4$ — weak separation
- PS-PMMA ($n \approx 10^3$): $\chi_c \approx 2 \cdot 10^{-3}$, $\chi_{\text{exp}} \approx 4 \cdot 10^{-2}$ — strong

Forcing a covalent bond $\rightarrow$ **di-block copolymer** PS-*b*-PMMA, modelled by Ohta-Kawasaki ^[5]:

$$F(u) + \frac{\gamma}{2} \iint (u - \bar{u}) G(x, y) (u - \bar{u}).$$



Frustration selects a mesoscale $\lambda \sim (\varepsilon/\gamma)^{1/3}$.

Gradient correction

At an interface φ varies sharply; use the **local fraction** $\varphi_\eta(x) = \frac{1}{\#(\Lambda_\delta \cap B_\eta(x))} \sum_{y \in \Lambda_\delta \cap B_\eta(x)} \sigma(y)$, with $\delta \ll \eta \ll \text{diam } \Omega$.

The bond energy between neighbours is

$$u(x, x + \delta e_i) = (U_{AA} + U_{BB} - 2U_{AB}) \varphi(x) \varphi(x + \delta e_i) + (U_{AB} - U_{BB}) (\varphi(x) + \varphi(x + \delta e_i)) + U_{BB}.$$

The **exact** identity $\varphi(x) \varphi(x + \delta e_i) = \frac{1}{2} [\varphi(x)^2 + \varphi(x + \delta e_i)^2] - \frac{1}{2} [\varphi(x + \delta e_i) - \varphi(x)]^2$ splits bulk + interface ^[6].

$$F_\delta(\varphi) = \delta^d \sum_{x \in \Lambda_\delta^0} \left[W(\varphi) + \frac{\chi}{2} \delta^2 \sum_{i=1}^d \left(\frac{\varphi(x + \delta e_i) - \varphi(x)}{\delta} \right)^2 \right].$$

The discrete Modica–Mortola functional — our rigorous starting point.

Discrete-to-continuum: two limits

As $\delta \rightarrow 0$ in L^2 , the bulk relaxes to its convex envelope ^[7]:

$$F_\delta \xrightarrow{\Gamma-L^2} \int_{\Omega} W^{**}(\varphi) \, dx.$$

Rescaling to a **surface** energy,

$$F_\delta^1 := \frac{F_\delta - \min F_0}{\delta} = \delta^{d-1} \sum_{x \in \Lambda_\delta^0} \left[W(\varphi) + \frac{\chi}{2} \sum_{i=1}^d (\varphi(x + \delta e_i) - \varphi(x))^2 \right].$$

Is the limit $\sigma \cdot \text{Per}(\{\varphi = 0\})$? **No** — it is anisotropic.

The anisotropic surface tension

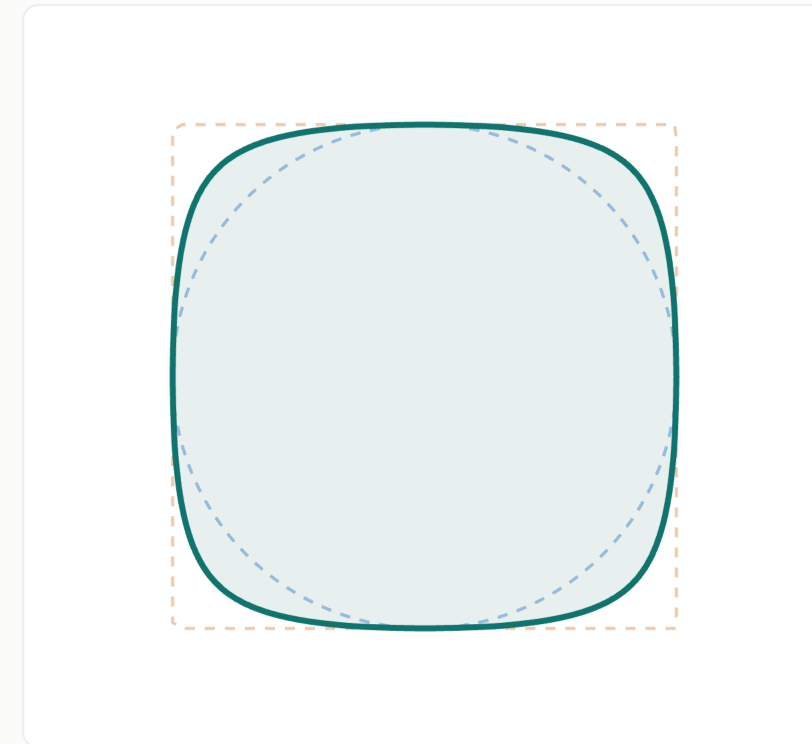
$$F_\delta^1 \xrightarrow{\Gamma-L^1} \int_{\partial^*\{\varphi=0\} \cap \Omega} \sigma_{\text{nn}}(\nu) \, d\mathcal{H}^{d-1}.$$

$$\sigma_{\text{nn}}(\nu) = \sup_{\{\lambda_i\} \in \Delta^{d-1}} \sum_{i=1}^d \gamma(\lambda_i) |\nu_i|,$$

with the optimal 1D transition cost ^[8]

$$\gamma(\lambda) = \inf_{\varphi_{\mp\infty} = \varphi_{1,2}} \sum_{k \in \mathbb{Z}} \left[\lambda W(\varphi_k) + \frac{\chi}{2} (\varphi_{k+1} - \varphi_k)^2 \right].$$

So $\sigma(\nu) = c\|\nu\|_2$? **No** — direction-dependent; the Wulff shape is **faceted**, interpolating disk $\rightarrow \ell^1$ -cube.



$$\chi = 3.7 \cdot \sigma(45^\circ) / \sigma(0^\circ) = 1.165$$

How to obtain isotropy? ($d = 2$)

Take bonds $b \in B$ with couplings J_b ^[8]: nearest $\{(1, 0), (0, 1)\}$ (C_1), next-nearest $\{(1, 1), (1, -1)\}$ (C_2), next-next-nearest $\{(2, 0), (0, 2)\}$ (C_3).

For a planar interface $\varphi(x) = \psi(x \cdot \nu)$ with $q = b \cdot \nu$:

$$(\Delta_b \varphi)^2 = q^2 \psi^{(1)} \psi^{(1)} + q^3 \psi^{(1)} \psi^{(2)} + q^4 \left(\frac{1}{4} \psi^{(2)} \psi^{(2)} + \frac{1}{3} \psi^{(1)} \psi^{(3)} \right) + \dots$$

Summing over bonds, the odd powers integrate to zero ($\int \psi^{(1)} \psi^{(3)} = -\int \psi^{(2)} \psi^{(2)}$), giving

$$\sigma(\nu) \approx \int_{\mathbb{R}} \left[W(\psi) + K_2(\nu) \psi^{(1)} \psi^{(1)} + K_4(\nu) \psi^{(2)} \psi^{(2)} + K_6(\nu) \psi^{(3)} \psi^{(3)} + \dots \right] dt$$

The even powers then give

$$K_{2k}(\nu) := \sum_{b \in B} J_b (b \cdot \nu)^{2k}, \quad K_2(\nu) = C_1 + 2C_2 + 4C_3 \quad (\text{isotropic gradient stiffness}).$$

The anisotropy lives in $K_4, K_6, \dots$ — the higher harmonics.

How to obtain isotropy? ($d = 2$)

Leading anisotropy:

$$K_4(\theta) := \left(\frac{3}{4}C_1 + 3C_2 + 12C_3\right) + \left(\frac{1}{4}C_1 - C_2 + 4C_3\right) \cos 4\theta;$$

With NN+NNN ($C_3 = 0$), choosing $C_2 = \frac{1}{4}C_1$ kills the $\cos 4\theta$ term – K_4 becomes **isotropic**.

But the next order then survives:

$$K_6(\theta) = \left(\frac{5}{8}C_1 + 5C_2\right) + \left(\frac{3}{8}C_1 - 3C_2\right) \cos 4\theta \xrightarrow{C_2=C_1/4} \frac{3}{8}C_1 (5 - \cos 4\theta).$$

One shell is not enough.

Higher anisotropy:

$$K_6(\theta) := \left(\frac{5}{8}C_1 + 5C_2 + 40C_3\right) + \left(\frac{3}{8}C_1 - 3C_2 + 24C_3\right) \cos 4\theta$$

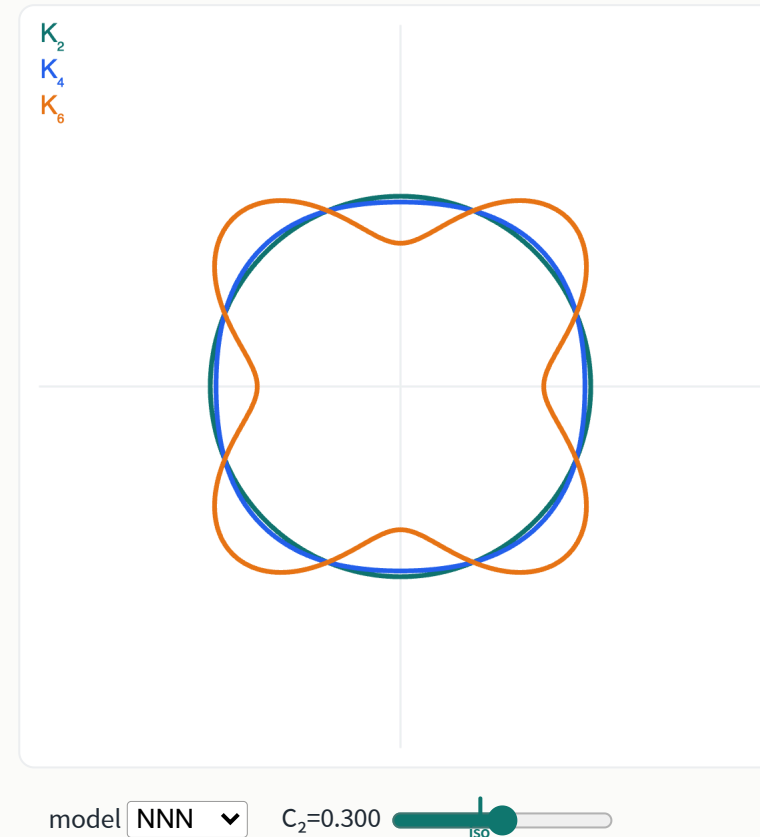
A hierarchy of harmonics

Each fine tuned shell removes one more order: [8]

- NN: K_2 isotropic, K_4 survives;
- NN+NNN ($C_2 = \frac{1}{4}C_1$): K_4 isotropic, K_6 survives;
- NN+NNN+NNNN ($C_2 = \frac{3}{8}C_1$, $C_3 = \frac{1}{32}C_1$): K_4, K_6 isotropic, K_8 survives;
- ... a new harmonic always reappears.

No finite range is *exactly* isotropic

⇒ one needs a full kernel J .



Non-local interactions

A Kac potential $J_\varepsilon(r) = \varepsilon^d J(\varepsilon r)$, with J even and isotropic and interaction range ε independent of δ ^[9]:

$$F_{\delta,\varepsilon}(\varphi) = \sum_{x \in \mathbb{T}_\delta^d} \delta^d \left[W(\varphi(x)) + \frac{1}{4} \sum_{y \in \mathbb{T}_\delta^d} J_\varepsilon(x - y) (\varphi(x) - \varphi(y))^2 \right].$$

$\delta \rightarrow 0$:

$$F_{\delta,\varepsilon} \xrightarrow{\Gamma} F_\varepsilon(\varphi) = \int_{\mathbb{T}^d} W(\varphi) + \frac{1}{4} \iint_{\mathbb{T}^d} J_\varepsilon(x - y) (\varphi(x) - \varphi(y))^2.$$

$\varepsilon \rightarrow 0$:

$$F_\varepsilon \xrightarrow{\Gamma} F_0(\varphi) = \int_{\partial^* \{\varphi=0\}} \sigma_0(\nu) d\mathcal{H}^{N-1}, \quad \sigma_0(\nu) = \inf \left\{ \frac{1}{|C|} F(\varphi, T_C) : \varphi \text{ } C\text{-periodic, } \varphi_{\mp\infty} = \varphi_{1,2} \right\}.$$

A kernel is like **infinitely many neighbour shells at once** — it cancels every harmonic K_{2k} , not just finitely many.

If J is isotropic $\Rightarrow F$ isotropic $\Rightarrow \sigma_0$ isotropic.

Isotropy at last.

The gradient is only the first order

The square-gradient is the **leading term** of the non-local penalization. Taylor-expanding $\varphi(y) = \varphi(x) + \nabla\varphi(x) \cdot (y - x) + \dots$,

$$\frac{1}{4} \iint J(x - y) (\varphi(x) - \varphi(y))^2 \approx \frac{\kappa}{2} \int_{\Omega} |\nabla\varphi|^2, \quad \kappa = \frac{1}{4d} \int_{\mathbb{R}^d} J(z) |z|^2 \, dz$$

— the gradient coefficient is the **second moment** of the kernel.

And the sharp-interface Γ -limit is **largely independent of the shape** of the penalization:

- Fonseca–Mantegazza ^[10]: $\int \frac{1}{\varepsilon} W(u) + \varepsilon^3 \|\nabla^2 u\|^2$
- Brusca–Donati–Solci ^[11]: $\int \frac{1}{\varepsilon} W(u) + \varepsilon^{2k-1} \|D^k u\|^2$

Same interfacial energy — the **universality** of Modica–Mortola.

Thank you for the attention!

- microscopic **lattice** model → Flory–Huggins free energy of mixing
- **entropy vs energy** → double well; transition at $\chi_c = 2$
- **gradient correction** → discrete **Modica–Mortola** functional
- Γ -limits: isotropic perimeter (decoupled) vs **anisotropic** (coupled)
- restoring isotropy: neighbour shells → **non-local kernel**
- **universality**: the Γ -limit barely depends on the penalization

ongoing Notes on phase separation — L. Pignatelli

Flory (1942) · Cahn–Hilliard (1958) · Lebowitz–Penrose (1966) · Modica–Mortola (1977) · Ohta–Kawasaki (1986) · Modica (1987) · Dal Maso (1993) · Alberti–Bellettini (1998) · Fonseca–Mantegazza (2000) · Brusca–Donati–Solci (2024)